Undergraduate Capstone Project Proposal
Course: Computer Science Capstone (Fall Semester)
Project Title: Prototype of an Improved FAA Aviation Medical Certification Information Portal for Pilots and Aviation Medical Examiners (AMEs)
Team: [Group of undergraduates – names/roles TBD]

Problem Statement
Pilots must obtain an initial and periodic FAA medical certificates (Class 1/2/3 or BasicMed) to be able to fly. To start the process, they submit an application via MedXPress (medxpress.faa.gov) and then schedule an AME exam. Pilots are encouraged to research their medical conditions and come prepared to the AME exam, with a preference for the pilots to use FAA resources on their website. The AMEs use current guidance from the FAA AME Guide and related resources in making the determination to issue a medical certificate or to defer the case to the FAA for further review.

The current FAA website (faa.gov/pilots/medical certification) and MedXPress are widely reported as confusing, with conflicting or hard-to-find information, poor usability, session timeouts, inability to edit submitted applications or auto-populate prior data, limited document attachment, and inadequate status tracking.

This compounds pilot distrust of the process, increases deferrals to FAA review (causing months-long delays), and hinders AMEs from issuing certificates on-site. Surveys show MedXPress satisfaction around 71% and FAA medical representative services as low as 30%, with common complaints including re-entry of history, unclear questions, missing documentation prompts, and poor communication.1(opens in new tab)

Objectives
Prototype a user-centered website that: (1) helps pilots “preflight” their medical (understand conditions, required docs, next steps, and medications); (2) provides AMEs with clear, concise, searchable, up-to-date guidance to reduce deferrals and keep more pilots flying; and (3) draws from international best practices for transparency and usability.

Proposed Solution & Key Requirements
A modern, responsive prototype (web app) inspired by but improving on the FAA website, incorporating:

- Pilot features: Guided medical history wizard with auto-save, previous-application recall, condition-specific checklists/docs (e.g., clinical progress notes), medication lookup, AME locator, status tracker, and “what to expect” flows.

- AME features: Searchable AME Guide excerpts, disposition tables, alerts for updates, document upload simulation, and decision-support tools.
- Shared: Clear navigation, no conflicting sources, mobile-friendly, accessibility (WCAG), and educational content to build trust. Non-functional: Fast load, secure (no real PII in prototype), intuitive UX.

International approaches inform design: Australia's CASA myCASA/MRS portal offers online applications, progress tracking, user manuals, and flowcharts; UK CAA Cellma (via Customer Portal) enables record viewing, tracking, and AME search; Canada's Transport Canada site provides structured apply/manage pages and CAME locators. EASA emphasizes Easy Access Rules and shared repositories (EAMR) for consistency.2(opens in new tab)3(opens in new tab)

Methodology & Deliverables
Research current FAA/MedXPress pain points and international sites; user interviews/surveys (pilots/AMEs); wireframes/prototypes (Figma + simple frontend); iterative testing. Deliverables: 1-page proposal (this), prototype site, user guide, presentation, and report. Timeline: Weeks 1–4 research/design; 5–10 build/test; 11–15 refine/demo.

This prototype aims to demonstrate how better information architecture can reduce confusion, deferrals, and distrust while supporting aviation safety.

Preflight your medical - AOPA
What does "Pre-Flight Your Exam" mean? | Federal Aviation Administration

Frequently Asked Questions | Federal Aviation Administration